Movie Success Prediction Report

DS3000

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Abstract

This project examined whether Rotten Tomatoes audience and critics scores can be predicted using box office revenue, MPAA ratings, and genre. Linear Regression models were used with each feature separately, but the resulting R^2 values were close to 0. This indicated that these predictors alone explained none of the variability in audience and critic ratings. We saw modest improvement when we shifted towards Polynomial Regression, with the predictors only explaining about 5% of the variability in audience and critic ratings. We then implemented interaction terms and dummy variables with Polynomial Regression, which produced the best results (audience: MSE ≈ 0.0208, $R^2 \approx 0.5144$). We finally tested a PCA-based regression that retained ~95% of predictor variance (24 components) but produced worse results. Overall, our findings suggest that polynomial and interaction-based feature structure preserves more predictive signals than dimensionality reduction of indicator features, and that improving predictive performance will likely require incorporating additional variables beyond box office, rating, and genre

Introduction

Movie success is often influenced by how well it resonates with both critics and audiences, yet these two perspectives don't always align. Understanding what drives the differences or similarities between audience and critic reception can provide insight into broader patterns in film evaluation. In this project, we aim to investigate whether certain genres tend to receive more favorable responses from critics or audiences, and whether there is a measurable correlation between genre, box office revenue, rating, audience score, and critics score. The question we are trying to answer is whether or not there is a correlation between genre, box office revenue, or rating and audience or critic score. By analyzing movies released from 2000-2025 using data from Rotten Tomatoes, our goal is to uncover trends that reveal how critical acclaim and popular opinion intersect or diverge across different film genres. The key questions are as follows:

1. Is there a correlation between the genre to audience score and the genre to critics score?
2. Is there a correlation between the box office revenue to audience score, and the box office revenue to critics score?
3. Is there a correlation between the rating to audience score and the rating to critics score?

Data Description

Summary of the data processing pipeline:

1. Scraped raw data from Rotten Tomatoes - used selenium to get the full list of movies (Rotten Tomatoes URLs alone could not load more than the first 5 pages) and beautifulsoup to get all the data from movies
2. Cleaned the data to prepare the data frame for visualization and analysis
3. Visualized using matplotlib and numpy

We constructed a Rotten Tomatoes movie dataset through webscraping, using Selenium to load the full list of movies and BeautifulSoup to extract attributes for each page. This approach was necessary because deeper pages were not directly accessible through static URLs alone. We then cleaned and formatted the dataset, saving the preprocessed dataset into a .csv file and importing it into Jupyter Notebook. The cleaning process involved removing invalid or inconsistent rows, separating the nested scores field into criticsScore and audienceScore columns, converting scores from percent strings into numeric proportions, standardizing boxOffice values by converting “M” and “K” formats into absolute dollars, and transforming multi-genre lists into binary indicators for each genre. After cleaning, we engineered model-ready predictors such as log(boxOffice), dummy-coded MPAA rating, and genre indicators to support modeling of audience and critic scores.

```
Head of cleaned Rotten Tomatoes data (csv_files/cleaned_data.csv):
                                                  url  \
0  https://www.rottentomatoes.com/m/the_long_walk...
1        https://www.rottentomatoes.com/m/the_roses
2    https://www.rottentomatoes.com/m/afterburn_2025
3      https://www.rottentomatoes.com/m/coyotes_2025
4  https://www.rottentomatoes.com/m/the_hand_that...

                           title rating  boxOffice  criticsScore  \
0                   The Long Walk      R   34800000          0.93
1                       The Roses      R   15300000          0.93
2                       Afterburn      R          0          0.79
3                         Coyotes      R          0          0.93
4  The Hand That Rocks The Cradle      R          0          0.45

   audienceScore  Action  Adventure  Animation  Anime  ...  House & Garden  \
0           0.61       0          0          0      0  ...               0
1           0.95       0          0          0      0  ...               0
2           0.78       1          1          0      0  ...               0
3           0.83       0          0          0      0  ...               0
4           0.30       0          0          0      0  ...               0

   Nature  News  Reality  Short  Soap  Special Interest  Talk Show  Travel  \
0       0     0        0      0     0                 0          0       0
1       0     0        0      0     0                 0          0       0
2       0     0        0      0     0                 0          0       0
3       0     0        0      0     0                 0          0       0
4       0     0        0      0     0                 0          0       0

   Variety
0        0
1        0
2        0
3        0
4        0
```

The example of cleaned data

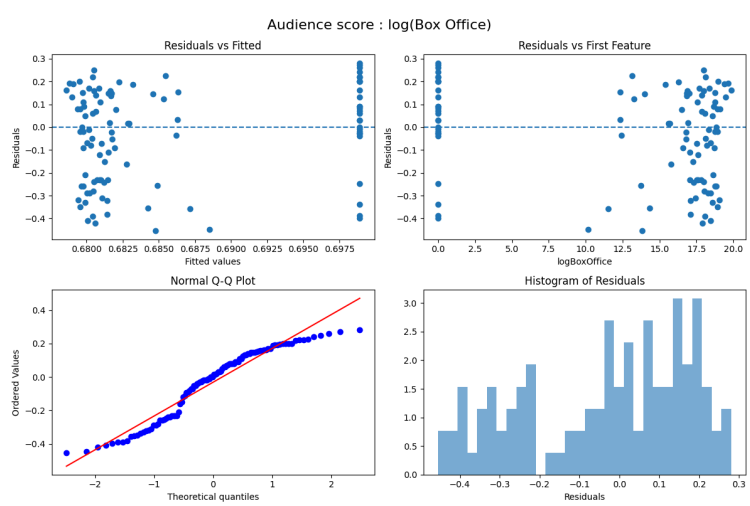
Method

First, we used linear regression to see whether features like box office revenue, movie rating (R, PG-13, etc), or genre membership have a linear relation to one of audience or critics scores. In order to do this, we made a model for linear regression, which uses the line of best fit for a linear relationship of predictors on the target variable, constant variance in the error terms, and residuals normally distributed. All three features had almost 0.0 R^2 values which meant that no feature had a meaningful variation which we used to figure out that the relationship between movie attributes and ratings is nonlinear. Because of that we made a model for polynomial regression, so that we can see if there's a non linear relationship while also being able to interpret the results. This new model's R2 value still remained low which meant that nonlinear effects were insufficient as well. In our final model with critics scores, movie ratings, and genre dummies, our final approach explicitly modeled how one feature's effect depends on the others. This model increased R^2 to about 0.51, showing that interactions explain substantially more of the variation in audience scores.

Results

Firstly, we analyzed **Model 1** using Linear Regression which revealed that log(Box Office) has almost no predictive power for either audience or critics scores, as reflected by R^2 values extremely close to zero (−0.0059 and 0.0085). This indicates that movies earning more at the box office do not systematically receive higher or lower ratings once the data is log-scaled to reduce skew. The residual plots would likely show no clear trend, supporting the conclusion that box-office performance alone is not a meaningful linear predictor of either score.

```
Model 1: log(Box Office) -> Audience score
Audience score : log(Box Office)
MSE = 0.04310705476493573
R^2 = -0.005926654067391901
```

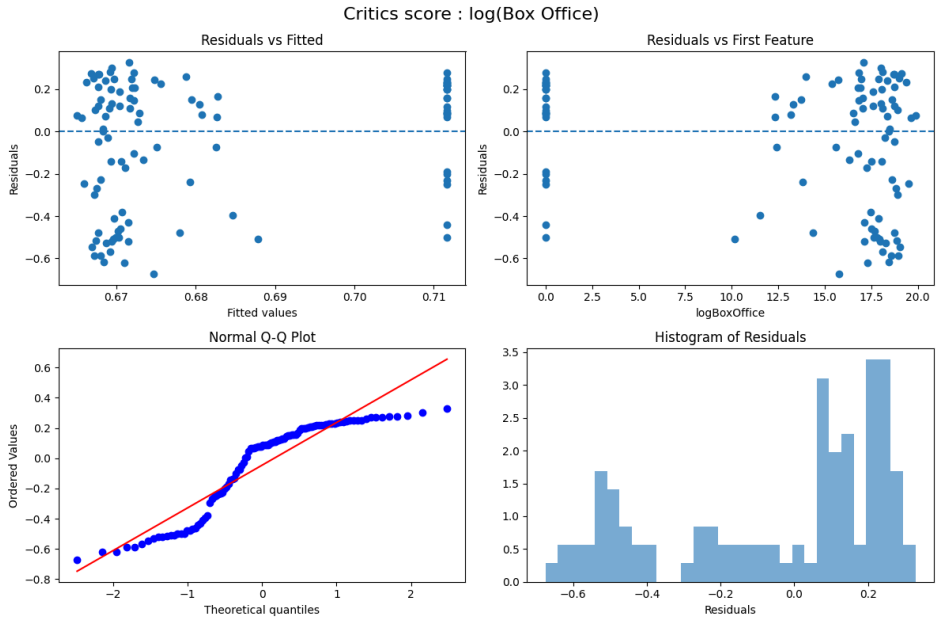


Model 1: log(Box Office) -> Critics score

Critics score : log(Box Office)

MSE = 0.09016801061870448

R^2 = 0.008519426348876147

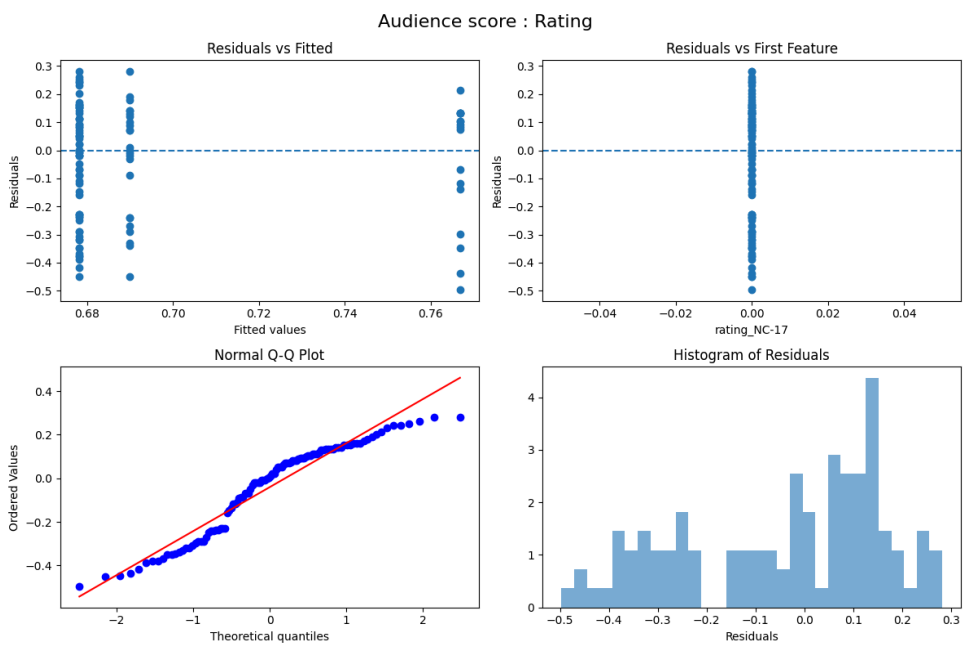


Model 2: Rating -> Audience score

Audience score : Rating

MSE = 0.04345808606723905

R^2 = -0.014118161126373874

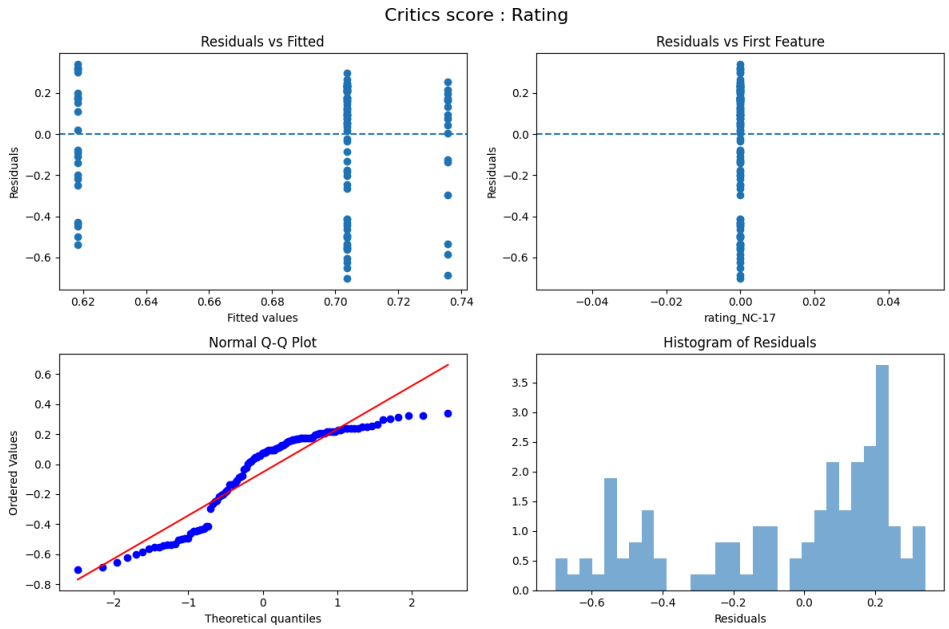


Model 2: Rating -> Critics score

Critics score : Rating

MSE = 0.09289655642650482

R^2 = -0.02148345542913277

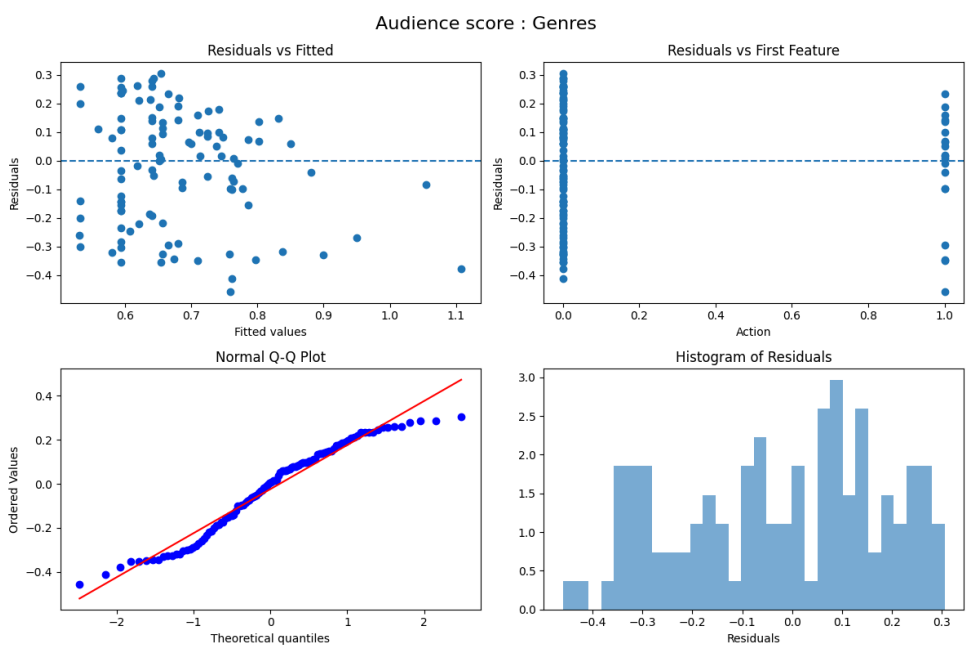


Model 3: Genres -> Audience score

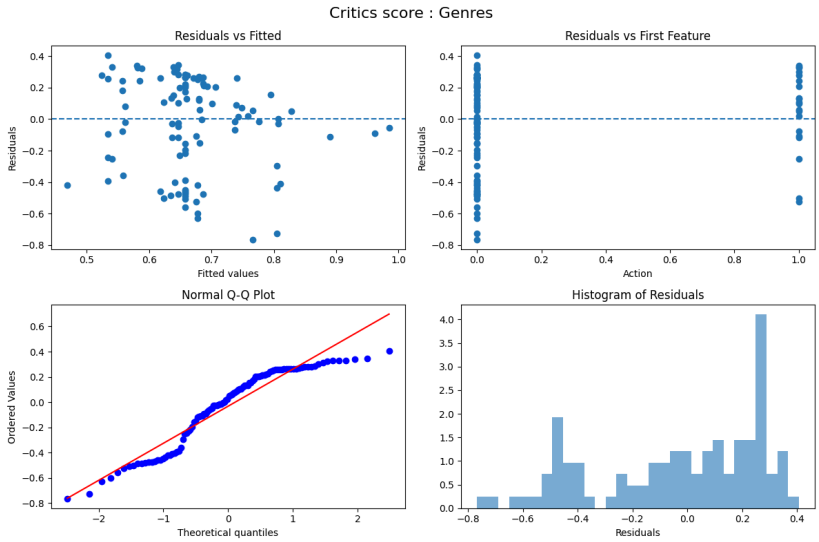
Audience score : Genres

MSE = 0.04080326938436746

R^2 = 0.04783343537043894



```
Model 3: Genres -> Critics score
Critics score : Genres
MSE = 0.09135201281714075
R^2 = -0.004499771600094293
```



Secondly, we analyzed **Model 2** using Polynomial regression. Similarly to Model 1, R^2 and MSE are both very close to 0 which shows that there isn't a significant correlation between audience score and critics score to box office and movie rating.

```
Audience score model:
MSE:  0.04078097954223351
R^2:  0.04835358100413534
Critics score model:
MSE:  0.08624749235187779
R^2:  0.0516291464539429
```

Then, we analyzed **Model 3: Polynomial Regression with Interaction Terms and Dummy Variables**, calculating MSE = 0.0208 and R^2 = 0.5144 showing significant improvement over Attempt 2's models by explaining approximately 51.4% of the variance in audience scores. Examining the Residuals vs Fitted plot, we observe residuals are reasonably scattered around y=0 with no strong systematic pattern, though there is slight heteroscedasticity as the spread appears wider at lower fitted values (0.5-0.7) compared to higher values, indicating minor violations of linearity and constant variance assumptions. The Residuals vs First Feature (Action) plot shows expected clustering at Action=0 and Action=1, with residuals randomly distributed within each group, supporting the independence assumption. The Q-Q plot reveals most points follow the diagonal line well, though deviations at both tails indicate minor normality violations, and the Histogram of Residuals shows an approximately bell-shaped

distribution centered near zero with slight asymmetry. The addition of polynomial terms (critics score up to degree 4) and interaction terms (critics×R-rating, critics×Action, R-rating×Action, and three-way interaction) dramatically improved model performance, with R^2 increasing from 0.05 to 0.5144 and MSE decreasing from 0.04-0.08 to 0.0208.

```
MSE:  0.020809202617256103
R^2:  0.5144058976719085
```

Lastly, we implemented PCA via sklearn library modules to predict the audience and critics score based on factors such as box office performance, Motion Picture Association ratings, and genre. The predictor variables were derived from log(boxOffice), MPAA rating (dummy coded), and Genre indicators, with audienceScore and criticsScore as the dependent variables. After standardizing the features, we set the PCA model to retain about 95% of the variance in the predictor space, which resulted in 24 components. While PCA was effective at reducing the dimensionality of the feature set, its predictive performance (audience $R^2 \approx 0.16$, critics $R^2 \approx 0.09$) was unsatisfactory in comparison to previous models. This suggests that polynomial and interaction-based feature structure may preserve more predictive signals than compressing categorical indicators, with model 3 remaining the most effective approach in this context.

```python
# preprocess the data
df_attempt4 = processed_df.copy()

# 1. Log transform boxOffice
df_attempt4['logBoxOffice'] = np.log1p(df_attempt4['boxOffice'])

# 2. Genre columns = everything after audienceScore in your file
genre_cols = df_attempt4.columns[df_attempt4.columns.get_loc('audienceScore') + 1:].tolist()
non_zero_genre_cols = [c for c in genre_cols if df_attempt4[c].sum() > 0]

# 3. One hot encode rating (only this, NOT url/title)
rating_dummies = pd.get_dummies(df_attempt4['rating'], prefix='rating', drop_first=True)


# seperate features and target variable
X = pd.concat(
    [
        df_attempt4[['logBoxOffice']],
        rating_dummies,
        df_attempt4[non_zero_genre_cols]
    ],
    axis=1
).fillna(0)

y = df_attempt4[["audienceScore", "criticsScore"]].copy()
```

```python
# standardize the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# apply PCA
pca = PCA(n_components=.95)
X_pca = pca.fit_transform(X_scaled)

# perform regression on principal components
model = LinearRegression()
model.fit(X_pca, y)

# predictions and evaluation
y_pred = model.predict(X_pca)
mse = mean_squared_error(y, y_pred, multioutput="raw_values")
r2 = r2_score(y, y_pred, multioutput="raw_values")

print(f'Components kept: {pca.n_components_}')
print(f'MSE for Audience and Critics Score: {mse}')
print(f'R^2 for Audience and Critics Score: {r2}')
```

```
Components kept: 24
MSE for Audience and Critics Score: [0.03825033 0.07178   ]
R^2 for Audience and Critics Score: [0.15782902 0.09437594]
```

Discussion

These findings highlight the complexity of addressing the movie-success prediction challenge and the limitations of simply using linear regression models. The best performing model was the Polynomial Regression with Interaction Terms and Dummy Variables, with values of MSE = 0.0208 and $R^2 = 0.5144$. This model demonstrates how implementing non-linear interactions, or dummy variables, between box office revenue, ratings, and genre can improve predictive power.

Conversely, Models 1 and 2 produced MSE and R2 values close to 0 and consistently violated key assumptions of linear regression. This suggests that when box office, ratings, and genre are linearly modeled on their own, they are not sufficient enough to predict how audiences actually respond to movies. Even our final model showed minor violations of the linearity, constant variance, and normality assumptions. Furthermore, about half of variance in audience scores is still unexplained, suggesting that some important features may be missing from our list.

One ethical implication is that if the model learns these patterns from historical rating data which have a certain impact on various genres, demographics, or rating categories, it can unintentionally amplify those biases when predicting future movie success because people will have expectations based on these results, which would lead to more people going to higher rated movies, making them rated even higher for variables like the box office revenue.

Future research discussions might include

- Exploring models with greater flexibility, such as ensemble models to combine multiple individual linear models into one
- Investigating even more additional features, such as production budget, cast popularity, director popularity, etc. that might increase audience score variability
- Implementing non-parametric models to account for the consistent violations on the assumptions of linear regression

These limitations and potential improvements serve as a building block for developing a more effective model in future iterations. Our analysis suggests that while implementing interaction terms and dummy variables into our polynomial regression model can substantially boost performance, accurately predicting movie success requires richer data and more sophisticated models than those explored in this preliminary study.